

KATAPLAY — GAME DESIGN DOCUMENT

Lead Game Designer: Duolingo, Lingokids, Pokémon, Candy Crush, Super Mario World **Versi:** 1.0 **Target:** Anak 4-10 tahun

1. CORE GAMEPLAY

Genre Definition

KataPlay adalah **Adventure Learning Game** — perpaduan antara:

Genre	Inspirasi	Persentase
Adventure / Exploration	Super Mario World	35%
Educational Puzzle	Duolingo / Lingokids	35%
Collection RPG	Pokémon	20%
Match / Candy Crush	Candy Crush (reward loop)	10%

Satu Kalimat Gameplay

"Anak menjelajahi dunia, menyelesaikan misi belajar, mengoleksi karakter, dan naik level — tanpa sadar sedang belajar Bahasa Indonesia."

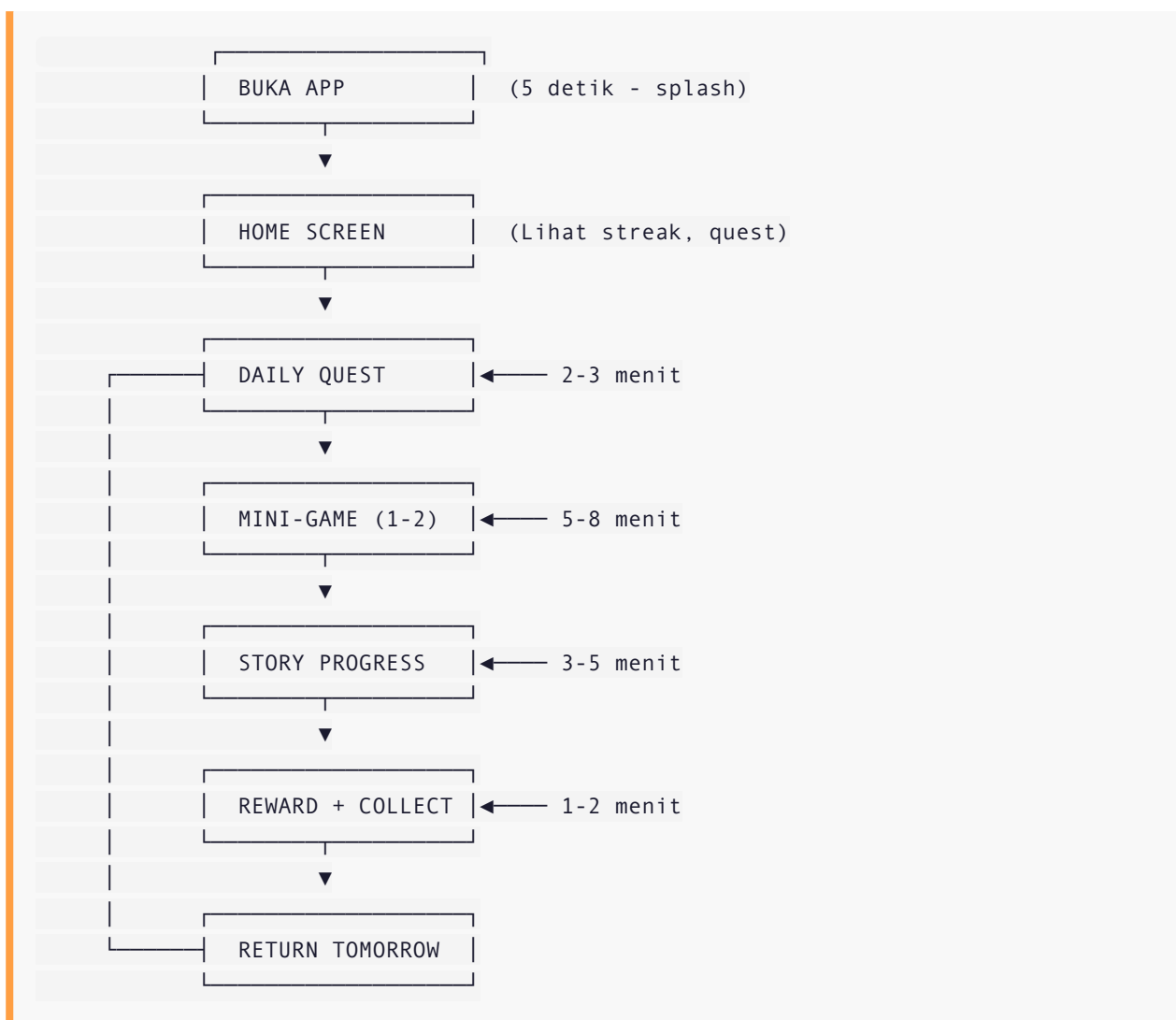
Core Interaction Model

EXPLORE → DISCOVER → PLAY → LEARN →
COLLECT → PROGRESS → REPEAT

Tahap	Aktivitas Anak	Mekanisme Game
Explore	Tap peta, pilih world, jalan ke node	Side-scroll atau point-and-click map
Discover	Temukan quest, item, karakter baru	Random encounter, hidden item
Play	Mainkan mini-game	Puzzle, drag-drop, match, tracing
Learn	Serap materi bahasa secara natural	Contextual learning in gameplay
Collect	Dapat bintang, koin, stiker, karakter	Reward screen + animation
Progress	Level naik, world baru terbuka	XP bar, level up celebration

2. CORE LOOP

The 15-Minute Daily Session



Time Budget per Session

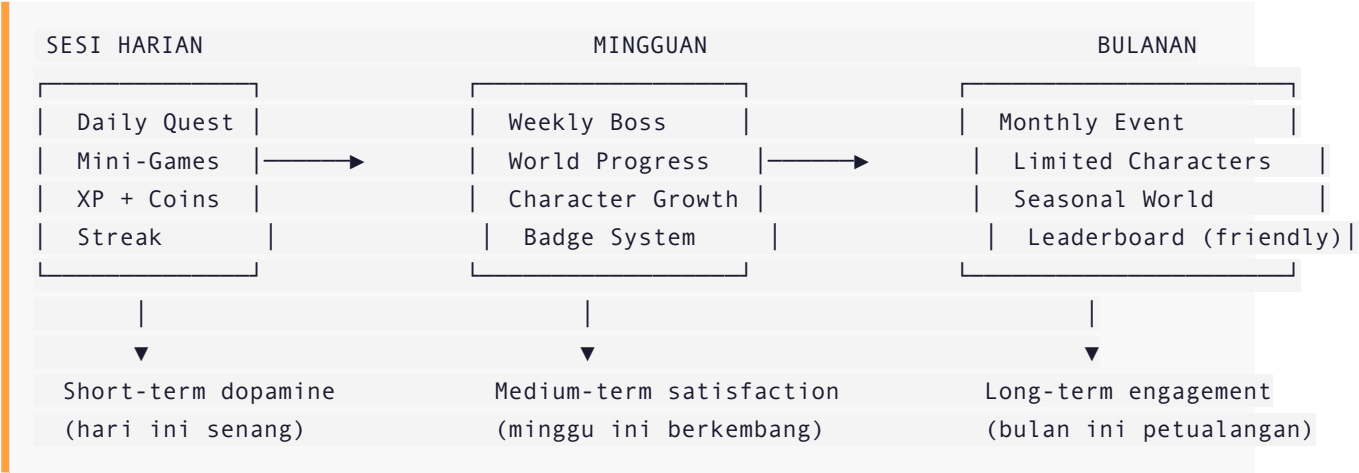
Aktivitas	Waktu	Persentase
Login + Streak	30 detik	3%
Daily Quest	2 menit	13%
Mini-Game (×2)	6 menit	40%
Story / Exploration	4 menit	27%
Reward + Collection	1.5 menit	10%
Navigation + Loading	1 menit	7%
Total	~15 menit	100%

Session Frequency Target

Usia	Recommended	Max
4-5 tahun	1 session/hari (15 menit)	20 menit
6-7 tahun	1-2 session/hari	30 menit
8-10 tahun	2 session/hari	40 menit

3. META LOOP

The Long-Term Progression



Meta Loop Drivers

Driver	Mekanisme	Frekuensi
Daily Streak	Login streak → reward	Harian
Quest Completion	Selesaikan quest → XP + item	Harian
Level Up	Kumpulkan XP → level baru	2-3 hari
World Unlock	Selesaikan world → world baru	1-2 minggu
Character Unlock	Koleksi bintang → karakter baru	1-4 minggu
Seasonal Event	Event khusus → limited items	Bulanan
Mastery	Kuasai semua mini-game → badge	Permanen

4. WORLD STRUCTURE

5 Worlds + Finale

World	Tema	Level	Boss	Skill Fokus
1. Pulau Huruf	Tropical island	10	Raksasa Huruf	Fonik, alfabet
2. Kebun Kata	Garden	12	Kumbang Kata	Kosakata dasar
3. Pantai Cerita	Beach	14	Raja Ombak	Kalimat sederhana
4. Gunung Kalimat	Mountain	16	Naga Kalimat	Struktur kalimat
5. Kerajaan Bahasa	Castle	20	Penyihir Kata	Membaca & menulis
Finale	Moon	5	Naga Terbang	Semua skill
Total		77	6 Bosses	

Detail world ada di dokumen terpisah: 02_World_Map_Design.md

5. PROGRESSION SYSTEM

Progression Pillars

Pillar	Apa yang Naik	Efek
Player Level	Level akun (1-100)	Unlock fitur, world, karakter
World Progress	Persentase per world	Unlock world berikutnya
Character Level	Level Zelby/Hazel/Alby	Skill pasif, costume
Mastery Score	Per kategori kata	Badge, achievement
Streak	Hari berturut-turut	Reward eksklusif

Player Level Table (1-30)

Level	XP Needed	Total XP	Unlock
1	0	0	Mulai
2	50	50	Daily Quest
3	80	130	Pulau Huruf Level 4
4	120	250	Mini-game baru
5	180	430	Karakter Hazel
6	240	670	-
7	320	990	World 2: Kebun Kata
8	400	1390	Mini-game baru
9	500	1890	-
10	600	2490	Karakter Alby
11	750	3240	Costume Zelby
12	900	4140	World 3
13	1100	5240	-
14	1300	6540	Mini-game baru
15	1500	8040	Boss Rush mode
16-20	+500/level	~13000	World 4
21-25	+750/level	~20740	Costume Hazel

Level	XP Needed	Total XP	Unlock
26-30	+1000/level	~30740	World 5

XP Sources

Sumber	XP	Frekuensi
Mini-game selesai	10-30	Per game
Mini-game perfect score	+10 bonus	Per game
Daily quest	50-100	Harian
Streak bonus	20-200	Harian
Boss defeated	100-300	Per world
Achievement	50-500	Sekali
First win of day	2x multiplier	Harian

6. REWARD ECONOMY

Currency Types

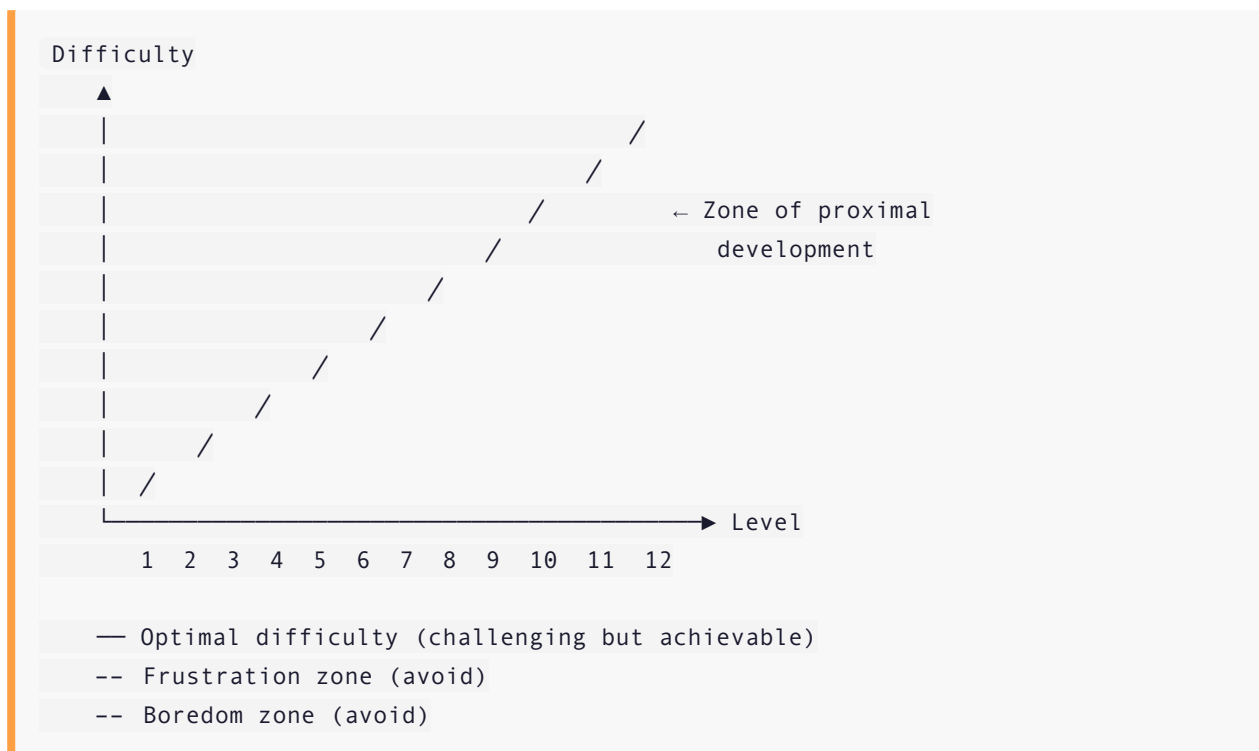
Currency	Icon	Fungsi	Didapat dari
Bintang	★	Unlock world, karakter	Mini-game, quest
Koin	🌐	Beli item, costume	Mini-game, streak
XP	📈	Naik level	Semua aktivitas
Stiker	📄	Koleksi, hiasan	Event, achievement
Badge	🏆	Prestasi permanen	Milestone

Detail ekonomi ada di dokumen terpisah: 05_Reward_Economy.md

7. DIFFICULTY CURVE

Curve Design Principle

"Easy to learn, hard to master — but never frustrating."



Difficulty Factors

Factor	Early Game (1-20)	Mid Game (21-60)	Late Game (61-100)
Word length	2-3 huruf	4-6 huruf	6+ huruf
Sentence length	2-3 kata	4-6 kata	7+ kata
Timer	No timer	30 detik	15 detik
Choices	2 pilihan	3-4 pilihan	4-5 pilihan
New mechanic	Setiap 3 level	Setiap 5 level	Setiap 8 level
Hint availability	Always	After 5s wrong	After 10s wrong

Adaptive Difficulty

Sistem akan mendeteksi dan menyesuaikan:

Skenario	Adjust
Anak menjawab benar 3x berturut-turut	Naik kesulitan +1
Anak salah 2x berturut-turut	Turun kesulitan -1
Anak idel 5 detik	Muncul hint
Anak main di atas 20 menit	Offer break + reward

8. LEARNING CURVE

Skill Progression

Level 1-20: FONIK & HURUF

└ Mengenal huruf A-Z, bunyi huruf, huruf kapital & kecil

Level 21-40: KOSAKATA DASAR

└ Kata benda (hewan, buah, warna, angka), kata kerja dasar

Level 41-60: KALIMAT SEDERHANA

└ S-P-O, kalimat positif, negatif, tanya

Level 61-80: MEMBACA & MENULIS

└ Paragraf pendek, cerita 2-3 kalimat, menulis kata

Level 81-100: MAHIR

└ Cerita panjang, pemahaman, menulis kalimat sendiri

9. PLAYER MOTIVATION

The 7 Motivators for Kids

#	Motivator	Mekanisme	Contoh
1	Rasa Ingin Tahu	Dunia tersembunyi, cliffhanger	"Apa yang ada di balik gunung?"
2	Collection	Karakter, stiker, badge	"Kumpulkan semua karakter!"
3	Achievement	Trophy, level up	"Level 10! Keren!"
4	Story	Narrative progression	"Lanjutkan cerita Zelby."
5	Mastery	Perfect score, 3 bintang	"Dapat 3 bintang di semua level!"
6	Social (light)	Bandingkan dengan teman	"Temanmu sudah di world 3!"
7	Autonomy	Pilih world, pilih game	"Mau main apa hari ini?"

What NOT to Use

✗ Jangan	Alasan
Competitive leaderboard	Anak bisa stres

✖ Jangan	Alasan
PvP (player vs player)	Tidak sesuai target usia
Pay-to-win	Tidak etis untuk anak
Loot box / gacha	Predatory mechanics
Hard fail state	Anak kecil mudah frustrasi
Time pressure (early)	Anxiety untuk pembaca pemula

10. SESSION DESIGN

Session Type 1: Daily Habit (15 min)

0:00 - Buka app, streak animation, daily reward
 0:30 - Daily quest muncul (3 misi kecil)
 2:00 - Mini-game 1: Review (materi kemarin)
 4:30 - Mini-game 2: New material
 7:00 - Story beat: cerita lanjutan
 10:00 - Mini-game 3: Challenge
 12:30 - Reward screen: XP, koin, item
 14:00 - "Sampai besok, ya!" dari Zelby

Session Type 2: Weekend Deep Dive (30 min)

0:00 - Buka app, streak bonus besar
 1:00 - Weekly boss try
 3:00 - Boss battle (5-7 menit)
 10:00 - Reward besar
 12:00 - World exploration (free play)
 18:00 - Side quest / achievement hunting
 25:00 - Shop / customization
 28:00 - "Petualangan seru banget!"

Session Type 3: Quick Check-in (5 min)

0:00 - Buka app
 0:30 - Streak reward
 1:00 - 1 mini-game cepat
 3:00 - Daily quest claim
 4:00 - "Besok main lagi, ya!"

11. RETENTION SYSTEM

Daily Retention Mechanics

Mekanisme	Psikologi	Day Impact
Streak counter	"Jangan putus!"	D1→D7 +35%
Daily reward escalation	"Besok lebih besar!"	D7→D30 +25%
Streak freeze item	"Aman kalau lupa sekali"	D30→D90 +20%
Zelby goodbye message	Emotional attachment	D1→D7 +15%

Weekly Retention Mechanics

Mekanisme	Psikologi	Retention Impact
Weekly boss	"Minggu ini harus kalahkan dia!"	W1→W2 +30%
Weekly quest reset	"Misi baru!"	W2→W4 +20%
Badge of the week	Limited time collection	W4→W8 +15%

Monthly Retention Mechanics

Mekanisme	Psikologi	Impact
Seasonal world	"Hanya bulan ini!"	M1→M2 +40%
Monthly leaderboard	Friendly competition	M2→M3 +15%
Character of the month	FOMO ringan	M3→M6 +25%

Retention Target Funnel

Metrik	Target
D1 Retention	70%
D7 Retention	45%
D30 Retention	30%
D90 Retention	20%
DAU/MAU	35%

Metrik	Target
Average session/week	5.5

12. CONTENT EXPANSION STRATEGY

Post-Launch Content Calendar

Bulan	Konten	Tipe
Month 1	Bug fixes, polish, balancing	Maintenance
Month 2	World 1 extension (+5 level)	World
Month 3	Seasonal event: Ramadhan	Event
Month 4	5 new mini-games	Gameplay
Month 5	World 2 extension (+5 level)	World
Month 6	Seasonal event: Kemerdekaan	Event
Month 7	Character costume shop	Monetization
Month 8	New character (surprise!)	Character
Month 9	Seasonal event: Natal & Tahun Baru	Event
Month 10	World 3 extension	World
Month 11	Mini-game creator contest	Community
Month 12	Anniversary event + major update	Event

Expansion Multiplier

Content Type	Dev Effort	Retention Impact	Monetization
New level	Rendah	Medium	Indirect
New mini-game	Medium	High	Indirect
New world	High	Very High	Conversion
Seasonal event	Medium	Very High	Direct
New character	High	High	Direct
Costume	Low	Medium	Direct
Badge/achievement	Low	Low	—

13. TECHNICAL DESIGN PRINCIPLES

Performance Targets

Metrik	Target	Notes
App size	<50 MB	Important for Indonesia
RAM usage	<256 MB	Mid-low end devices
Load time	<3 detik	Splash during load
Offline mode	Full gameplay	Sync when online
Animation FPS	24 fps	Cutscene, 12 fps idle
Battery usage	<10% per 30 min	Optimized

Save System

Data	Frequency	Priority
Player progress	Real-time	Critical
Game state	Every mini-game end	Critical
Inventory	Every transaction	Critical
Settings	On change	Normal
Analytics	Batch every 5 min	Background
Streak data	On login	Critical

14. GAME DESIGN RULES

Golden Rules of KataPlay

#	Rule	Why
1	No fail state	Anak kecil tidak boleh merasa gagal. Selalu "coba lagi."
2	Every action has reward	Setiap tap, swipe, drag — ada feedback positif.
3	3-click maximum	Dari home ke gameplay maksimal 3 tap.
4	Reading optional	Semua teks bisa didengarkan.

#	Rule	Why
5	No text walls	Maks 2 kalimat per layar.
6	Progress visible	Anak selalu tahu di mana posisinya.
7	Autonomy first	Anak bisa memilih apa yang ingin dimainkan.
8	Surprise & delight	Hidden easter eggs, random sparkles, funny sounds.

END OF GAME DESIGN DOCUMENT — CONFIDENTIAL